

Reading 3

Product Backlog and Workflow Management



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Program Management
& Scrum Master



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Agile Coach & Scrum Master



Topic 7

Product Backlog and Sprint

Outline



Sprint Goal



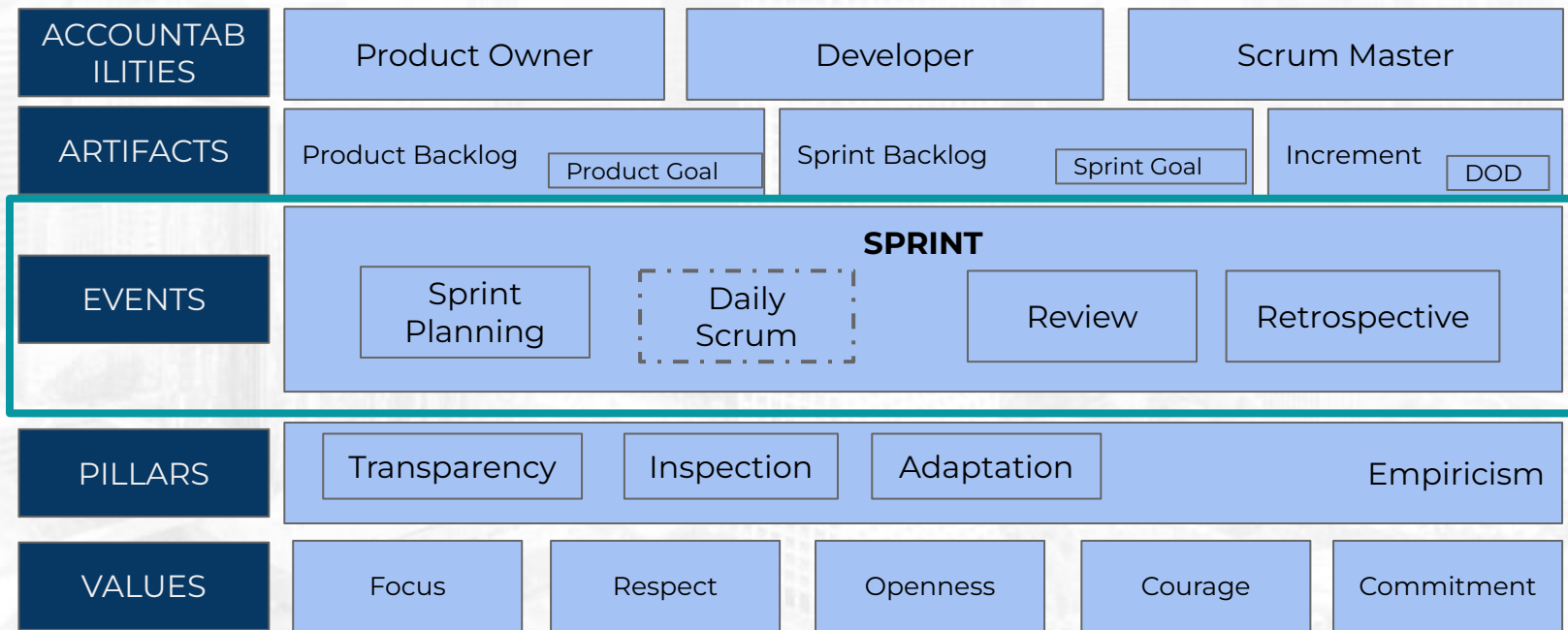
Sprint Planning



Sprint Backlog

The Sprint & Sprint Goal

There are 5 Scrum Events



Sprint

“The HeartBeat of Scrum”



A Container where Ideas turned into Value

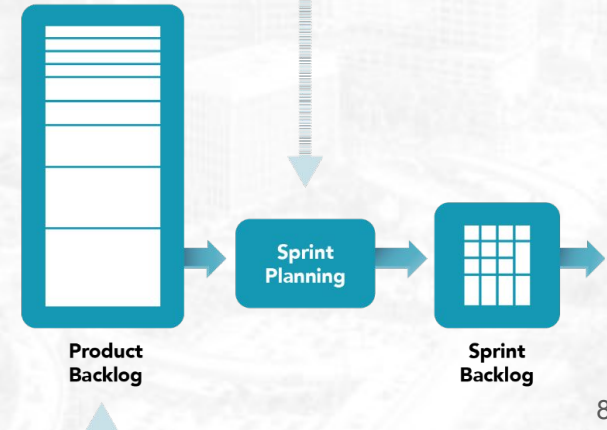
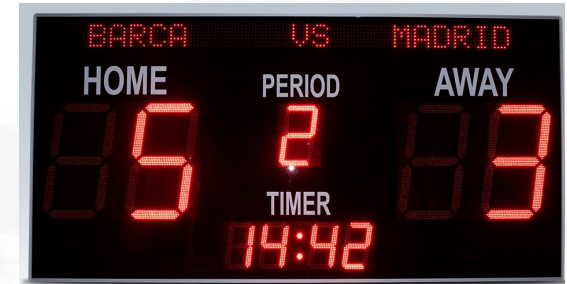
- ✓ TimeBox: 1 month calendar or less
- ✓ NO Changes that endanger Sprint Goal
- ✓ Quality does not decrease
- ✓ Product Backlog refined as needed & may be clarified &/ renegotiated with PO as more is learned
- ✓ It can be canceled by PO if Sprint Goal become obsolete

Sprint Goal

The Sprint Goal is the single objective for the Sprint.

Although the Sprint Goal is a commitment by the Developers, it provides flexibility in terms of the exact work needed to achieve it.

The Sprint Goal also creates coherence and focus, encouraging the Scrum Team to work together rather than on separate initiatives.



Outline



Sprint Goal



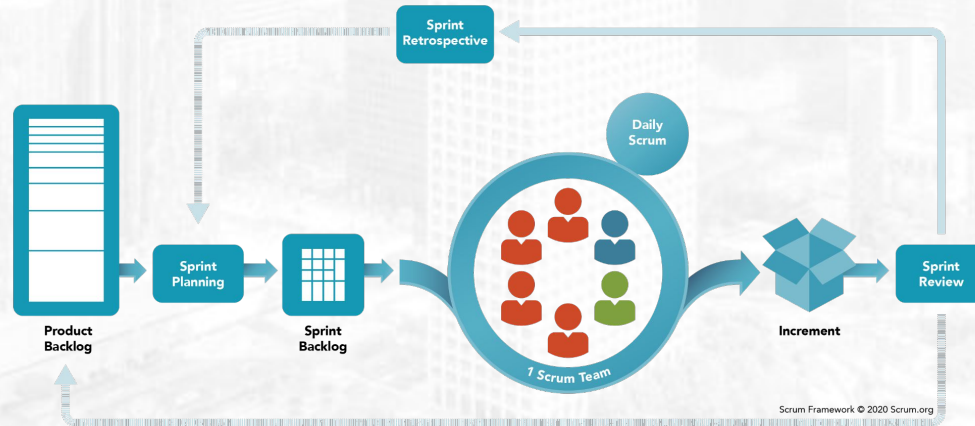
Sprint Planning



Sprint Backlog

The Sprint Planning

Scrum Event Map & Flow

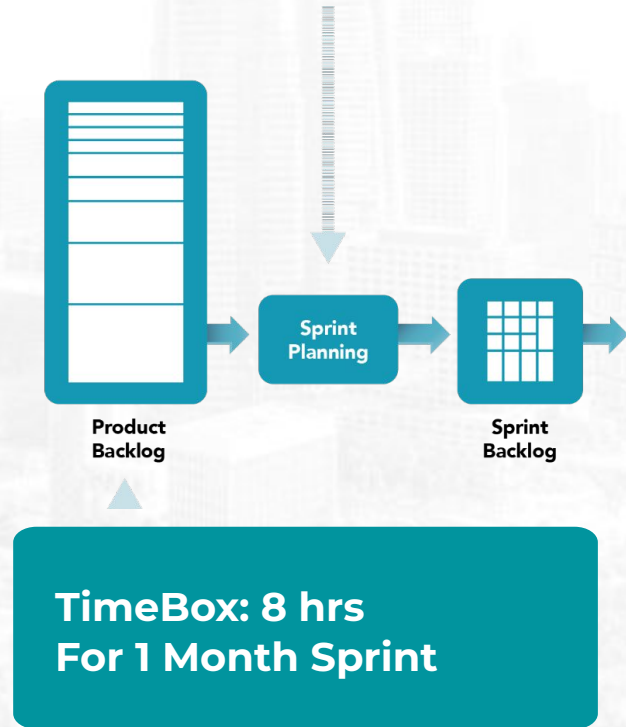


Sprint Planning Event

1 Why is this Sprint valuable?

2 What can be Done In this Sprint?

3 How will the chosen work get done?



Product Backlog Items Estimation

0	1	2	3	5	8	13
0	s	m	l	xl	xxl	~
NFC	1	TFB				

"work expands so as to **fill the time available** for its completion."
Parkinson's Law

A faded, grayscale background image of a city skyline with numerous skyscrapers, likely New York City, viewed from an elevated perspective.

Let's Exercise!

Outline



Sprint Goal



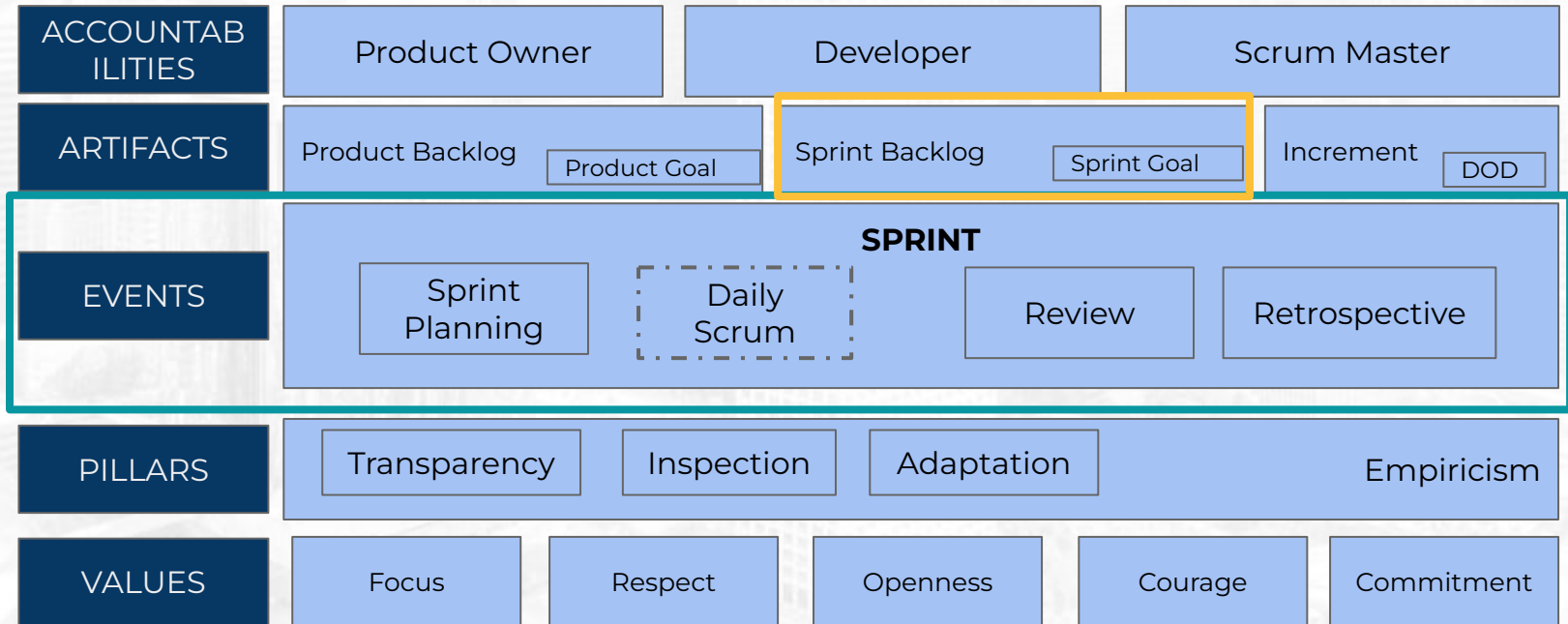
Sprint Planning



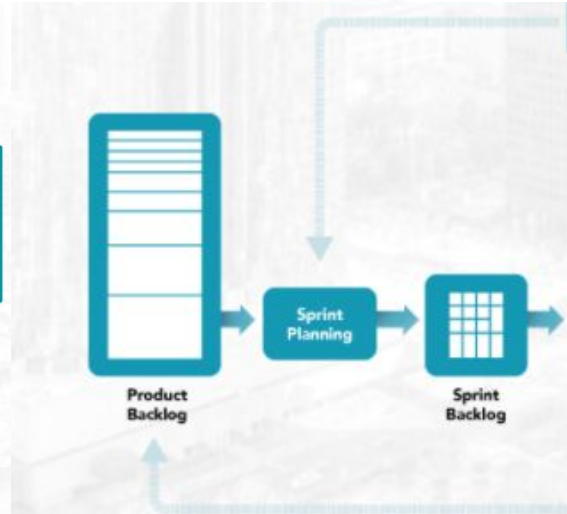
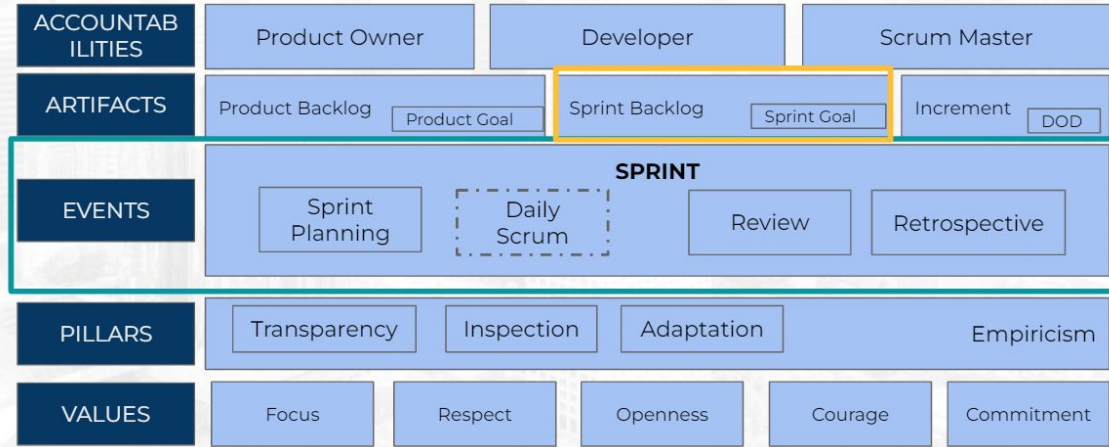
Sprint Backlog

Composing Sprint Backlog

There are 5 Scrum Events



Sprint Backlogs Components



The Sprint Goal,
the Product Backlog items selected for the Sprint,
plus the plan for delivering them
are together referred to as the Sprint Backlog.

Sprint Goal as the main Commitment of Sprint Backlogs

Sprint Backlog Examples

▼ PRAC Sprint 1 1 Mar – 15 Mar (7 issues) 38 6 3 Complete sprint

Sprint Goal: Increase App Rating by 25%

Item	Assignees	Status
PRAC-3 Product Backlog item 1	3	DONE
PRAC-4 Product Backlog item 2	5	IN PROGRESS
PRAC-5 Product Backlog item 3	1	IN PROGRESS
PRAC-6 Product Backlog item 4	11	TO DO
PRAC-7 Product Backlog item 5	1	TO DO
PRAC-8 Product Backlog item 6	21	TO DO
PRAC-9 Product Backlog item 7	5	TO DO

+ Create issue

Add epic / PRAC-4 1 Share More

Product Backlog item 2

In Progress

Description

Add a description...

Child issues

Order by ▼ More +

0% Done

Item	Description	Status
PRAC-21	How to 2.1	TO DO
PRAC-22	How to 2.2	TO DO

PRAC Sprint 1 9 days remaining Complete sprint

Sprint Goal: Increase App Rating by 25%

GROUP BY Subtask Insights

Item	Status	Subtasks
PRAC-3 Product Backlog item 1	DONE	- How to 1.1 PRAC-19 ✓ - How to 1.2 PRAC-20 ✓
PRAC-4 Product Backlog item 2	IN PROGRESS	- How to 2.2 PRAC-22 - How to 2.1 PRAC-21
PRAC-5 Product Backlog item 3	IN PROGRESS	
PRAC-6 Product Backlog item 4	TO DO	

List of PBI	Selected PBI	Work In Progress	Done (Increment)	Released
Product Goal PBI 7 PBI 8 PBI 9 PBI 10 + Add a card	Sprint Goal PBI 4 0/3 PBI 5 0/3 + Add a card	PBI 2 2/3 PBI 3 1/3 + Add a card	Definition of Done PBI 6 + Add a card	PBI 1 + Add a card

Outline



Sprint Goal



Sprint Planning



**Sprint Backlog
Empiricism**

Topic 8

Scrum Daily Operations

Outline

1

Increment &
DOD



2

Daily Scrum

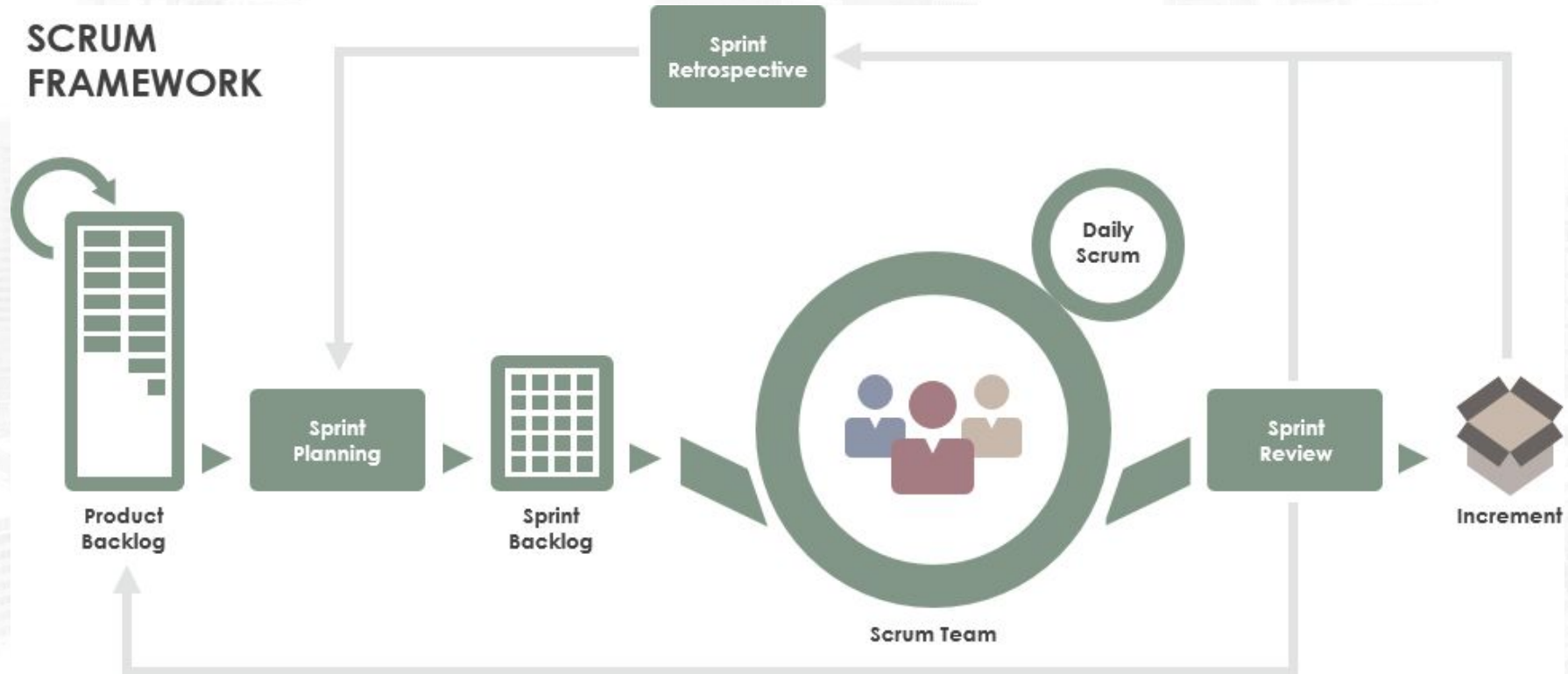


3

Sprint Backlog
Empiricism

Increment & DoD

SCRUM FRAMEWORK

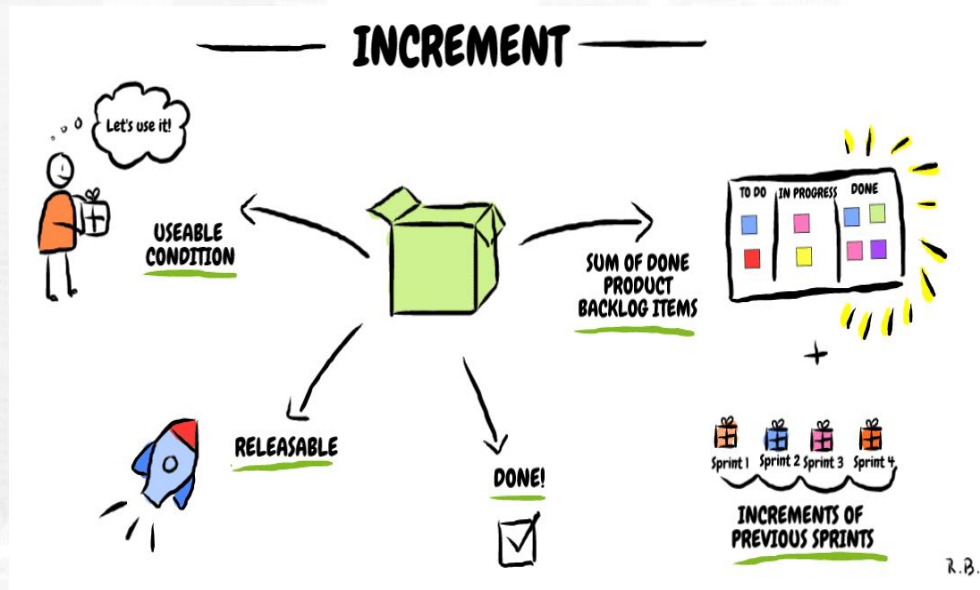


Artifact Increment



What is an Increment ?

An Increment is a concrete stepping stone toward the Product Goal. Each Increment is additive to all prior Increments and thoroughly verified, ensuring that all Increments work together. In order to provide value, the Increment must be usable.



The moment a Product Backlog item meets the Definition of Done, an Increment is born.

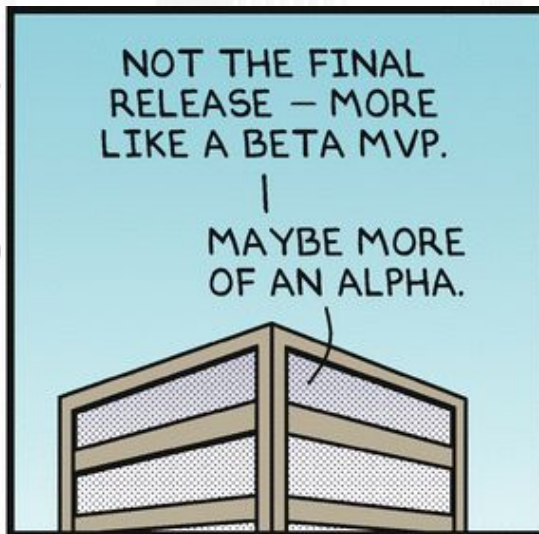
Then, How do you know when it is Done ?

Almost Done



@ScottAdamsSays

Dilbert.com



© 2016 Scott Adams, Inc. / Dist. by Universal Uclick



Definition of Done (DOD)



What is an DoD ?

The Definition of Done is a formal description of the state of the Increment when it meets the quality measures required for the product.

Scrum Guide 2020



Definition of Done (DoD) ?

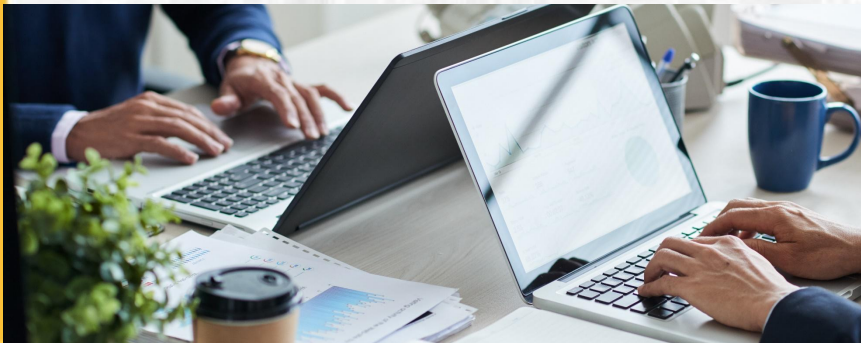
The Definition of Done is the commitment contained within the Increment artifact.

Who create the DoD ?

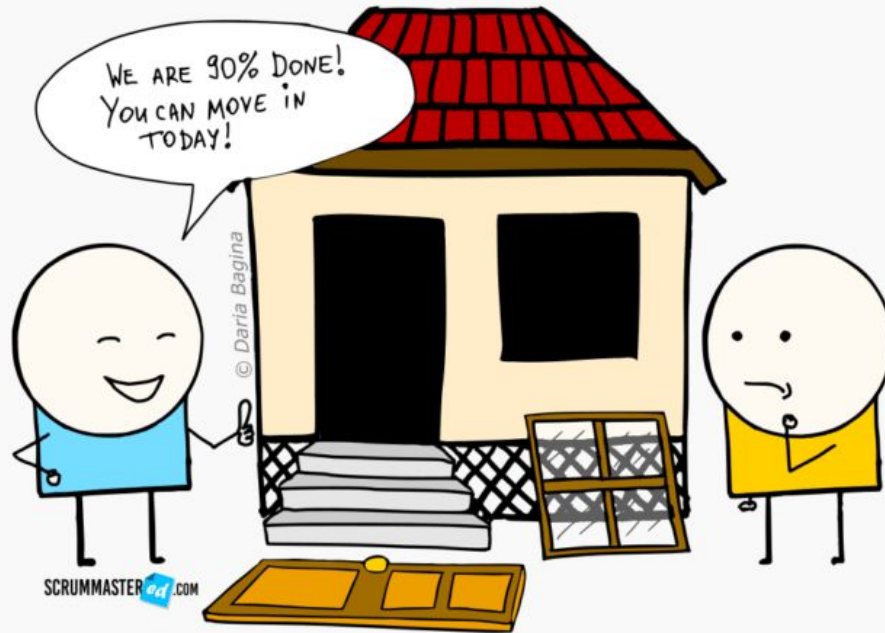
- ✓ The Scrum Team
- ✓ The standards of the organization

Notes

If a Product Backlog item does not meet the Definition of Done, it cannot be released or even presented at the Sprint Review. Instead, it returns to the Product Backlog for future consideration.



Definition of Done Examples



Define your DoD

how do we get started with creating a Definition of Done?

1

Transparent about all the activities that need to be done

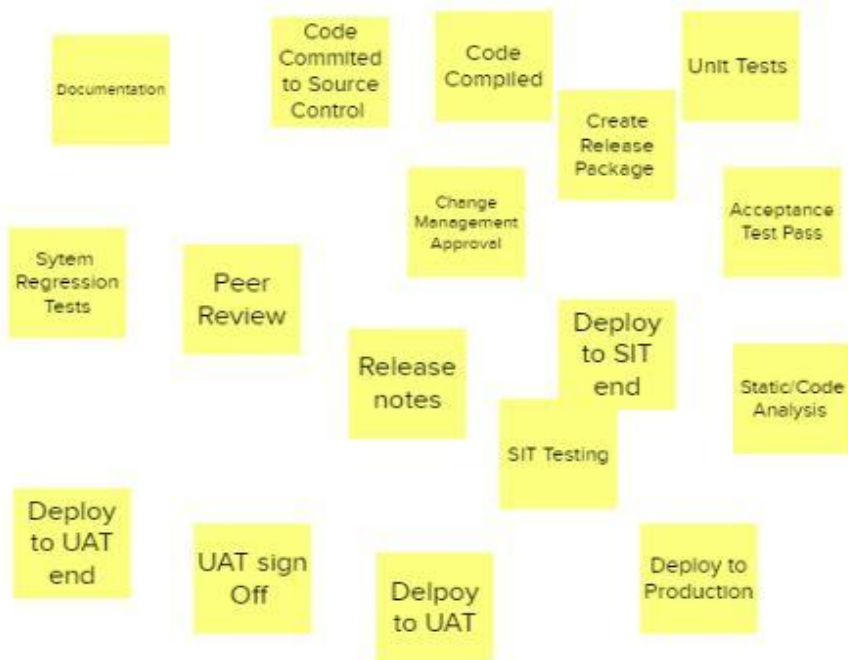
2

Benchmark. What is the current Definition of Done and what is currently not included?

3

Continuously Improve towards a Definition of Done that is closer to Releasable

List ALL Activities that need to be done before a Product Backlog Item (i.e. feature or bug) is consider Done, so than the increment can go Live!



Sky's the Limit!

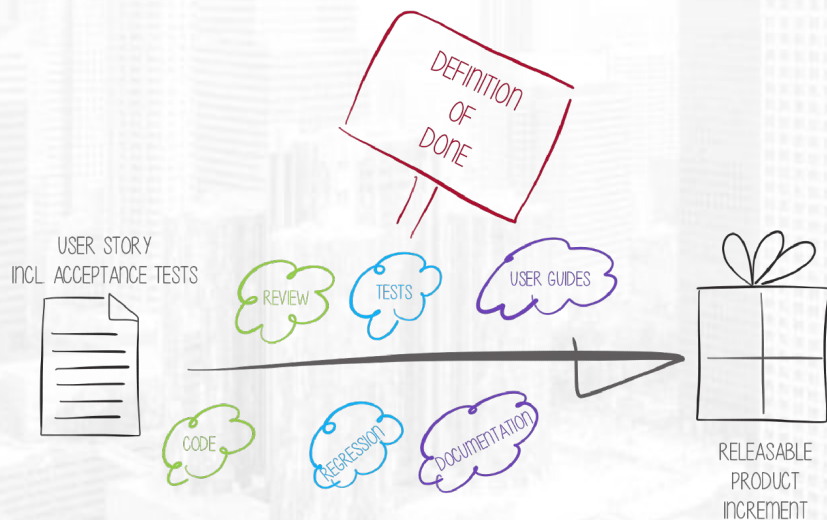
10 mins

1-2-4-ALL



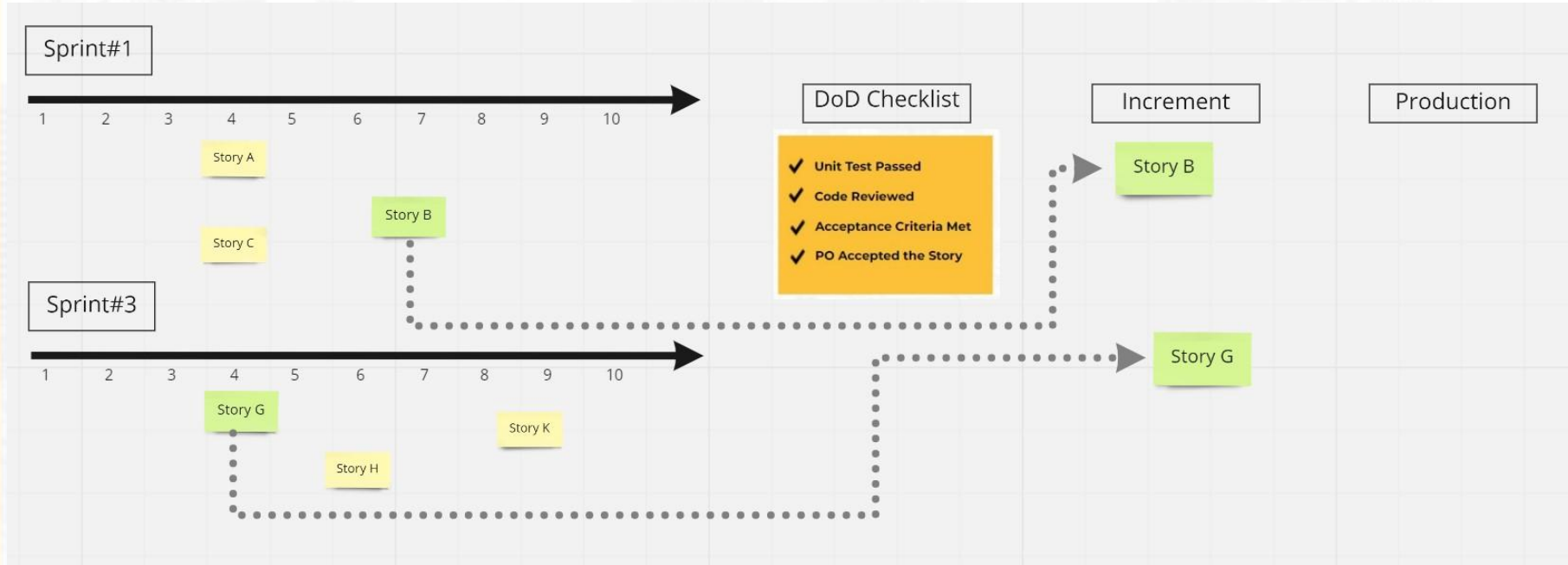
Engage Everyone
Simultaneously in
Generating Ideas,
and Suggestions

Definition of Done Examples

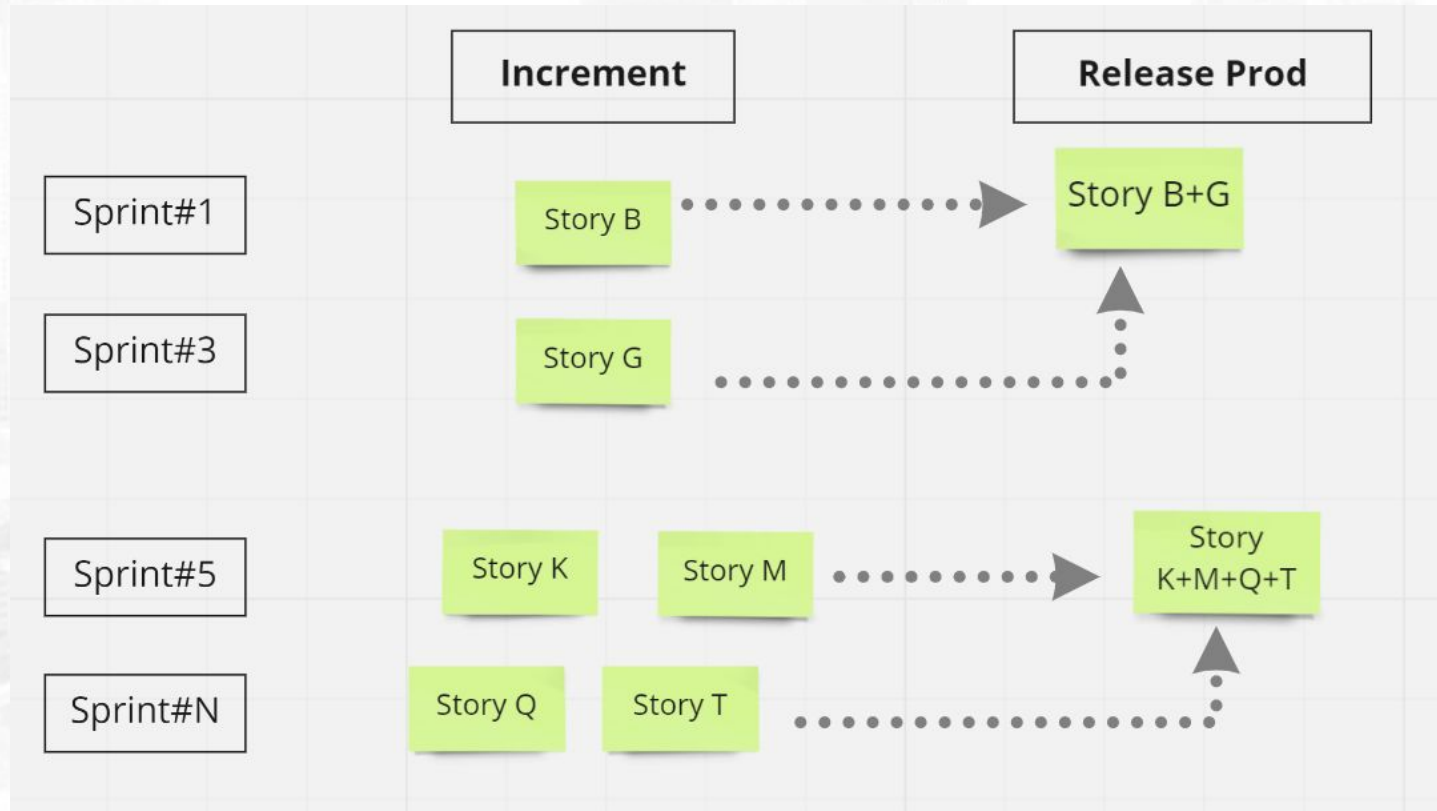


- ✓ **Unit Test Passed**
- ✓ **Code Reviewed**
- ✓ **Acceptance Criteria Met**
- ✓ **Documentations**

Increment



Increment



The Definition of Done creates transparency

The Definition of Done creates transparency by providing everyone a shared understanding of what work was completed as part of the Increment. If a Product Backlog item does not meet the Definition of Done, it cannot be released or even presented at the Sprint Review.

Outline



**Increment &
DOD**



Daily Scrum



**Sprint Backlog
Empiricism**

Daily Scrum

Events

Daily Scrum



What is Daily Scrum ?

The Daily Scrum is a 15-minute event for the Developers of the Scrum Team. Daily Scrums improve communications, identify impediments, promote quick decision-making, and consequently eliminate the need for other meetings.

Scrum Guide 2020



Daily Scrum

✓ **Who Attend ?**

The Developers. If the Product Owner or Scrum Master are actively working on items in the Sprint Backlog, they participate as Developers.

✓ **When/Where it is done ?**

It is held at the same time and place every working day of the Sprint

✓ **Timebox ?**

15 minutes



Daily Scrum **Anti-Patterns**



- ✖ Status Report
- ✖ Solving Problem
- ✖ Cluelessness
- ✖ Planning Meetings

The Developers use the Daily Scrum to inspect progress toward the Sprint Goal and to inspect how progress is trending toward completing the work in the **Sprint Backlog**

Daily Scrum is about the progress towards the **Sprint Goal**.
Where are we now? What is the next best thing we'd better take up today to bring us closer to our goal?
The Daily Scrum should be treated as an opportunity to **inspect & adapt**.

Outline



**Increment &
DOD**



Daily Scrum



**Sprint Backlog
Empiricism**

The Sprint Backlog Empiricism

Artifacts

Sprint Backlog



Inspecting

The Sprint Backlog is the set of Product Backlog items selected for the Sprint, plus a plan for delivering the product Increment and realizing the Sprint Goal. The Sprint Backlog is a forecast by the Development Team about what functionality will be in the next Increment and the work needed to deliver that functionality into a “Done” Increment

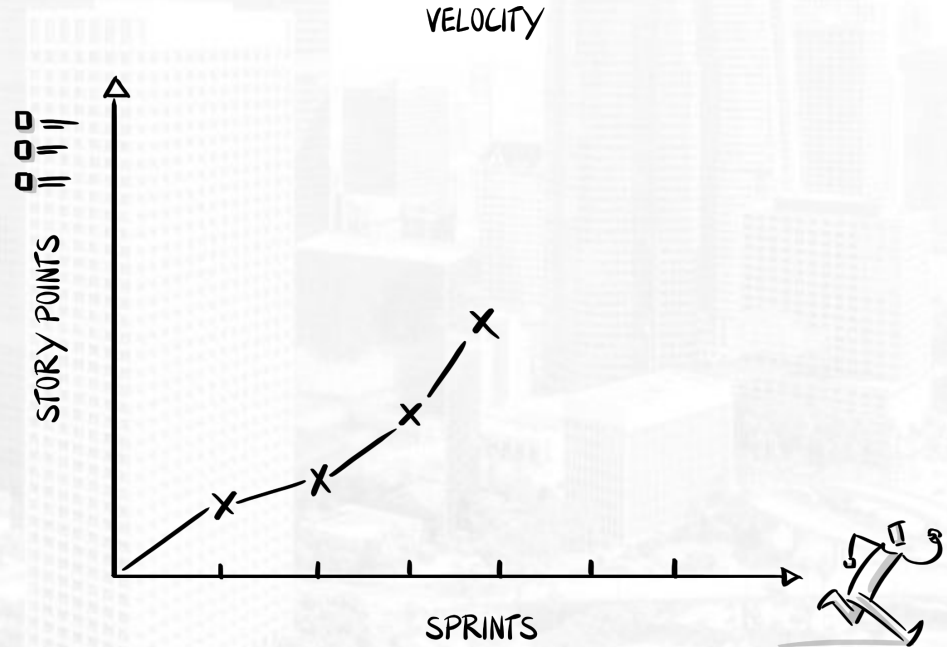


VELOCITY

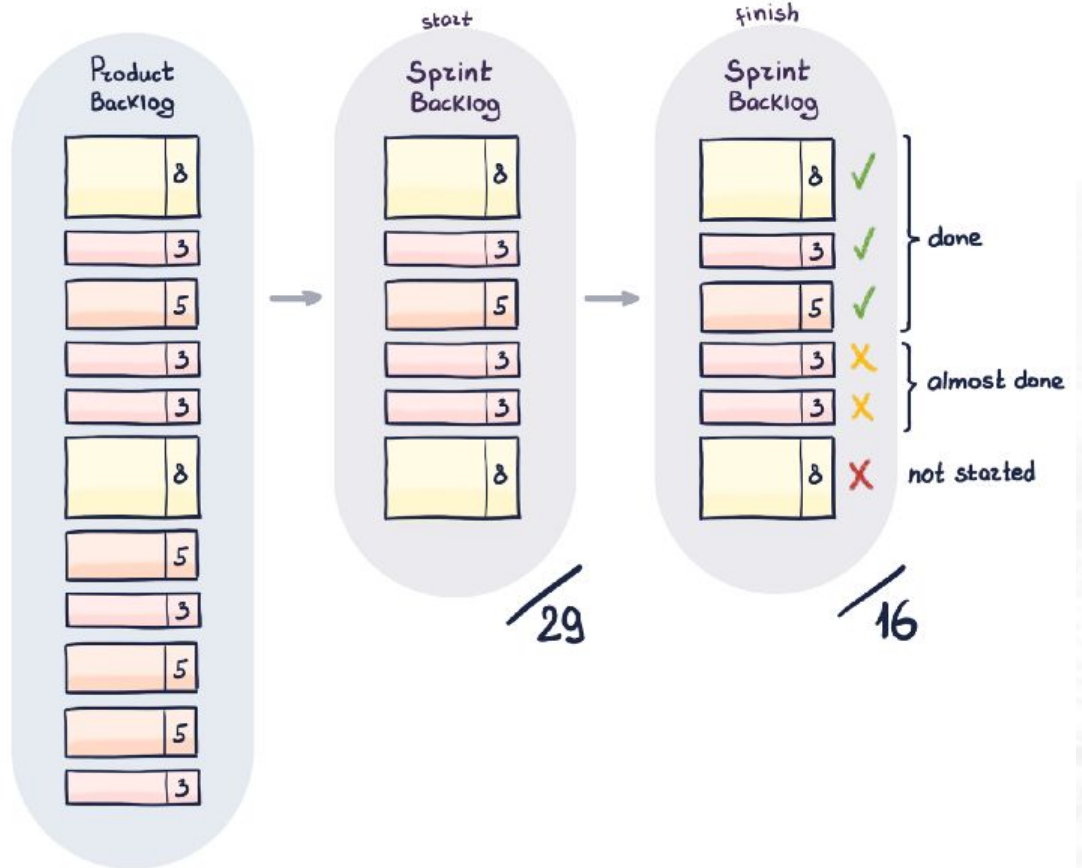


What is Velocity

Velocity is an indication of the ability to turn Product Backlog into releasable functionality across time, or for a specified price



Velocity



When is Velocity Useful ?

- As a guide when forecasting how much work can be done within a Sprint
- Creating **burn-down** and **burn-up** charts
- Creating forecasts for Release and Product delivery date

When is Velocity Not Useful ?

- Comparing performance of different Scrum Teams
- Assessing the performance of the Scrum Team to which it relates
- When used as a target rather than a measure

Burndown vs Burnup Chart



Monitoring Progress Towards Goals

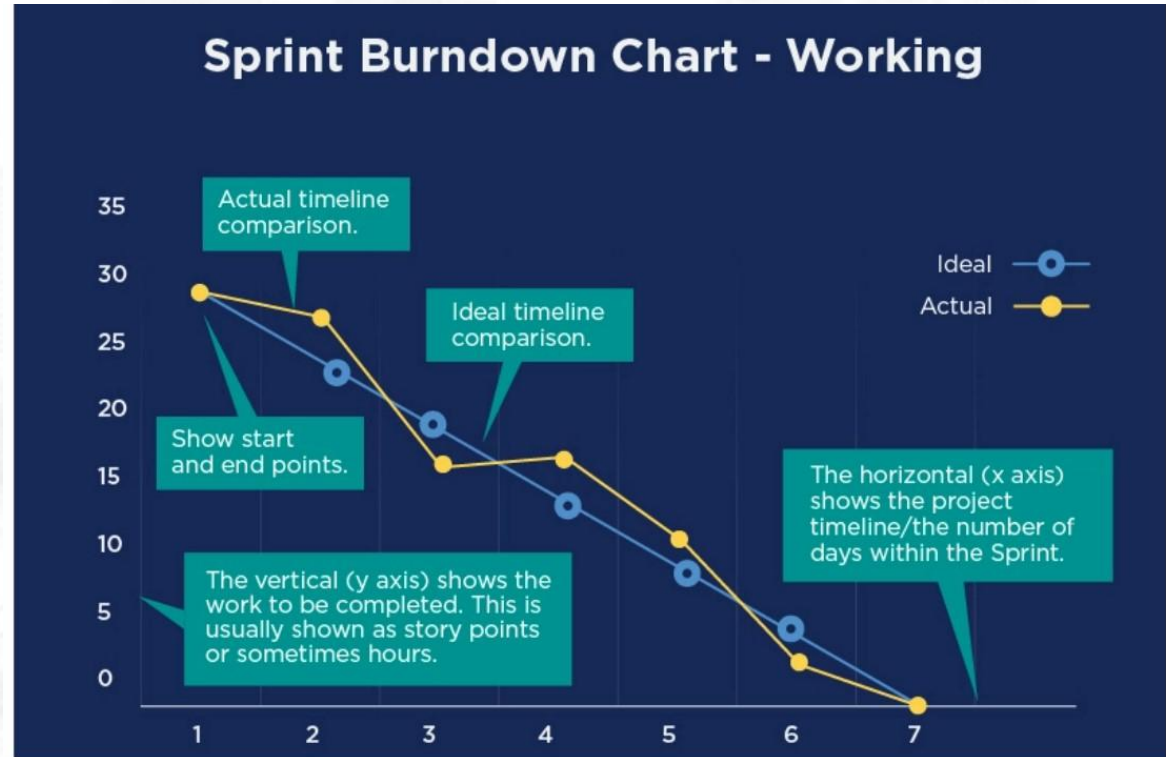
At any point in time, the total work remaining to reach a goal can be summed. The Product Owner tracks this total work remaining at least every Sprint Review. The Product Owner compares this amount with work remaining at previous Sprint Reviews to assess progress toward completing projected work by the desired time for the goal. This information is made transparent to all stakeholders.

Burndown Chart



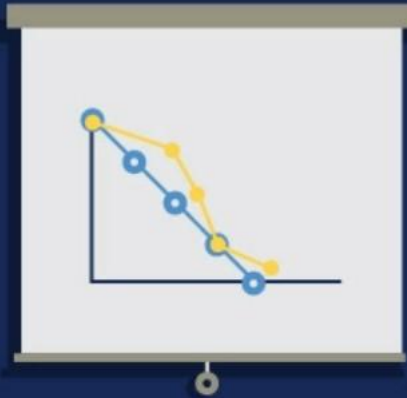
Why use Burndowns ?

- Make the reality of the project clear
- Show the impact of decisions
- Warn you early if things aren't going according to plan
- Get rid of all the wishful thinking around dates

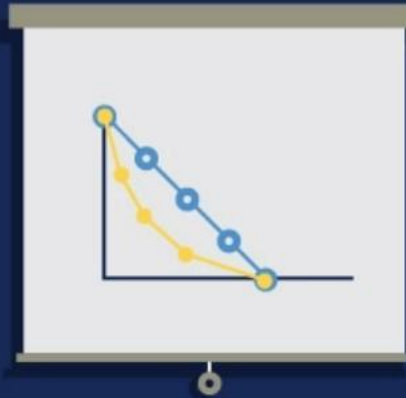


Burndown Chart

Burndown Chart - Interpretation



Completed work on time and met the goal.

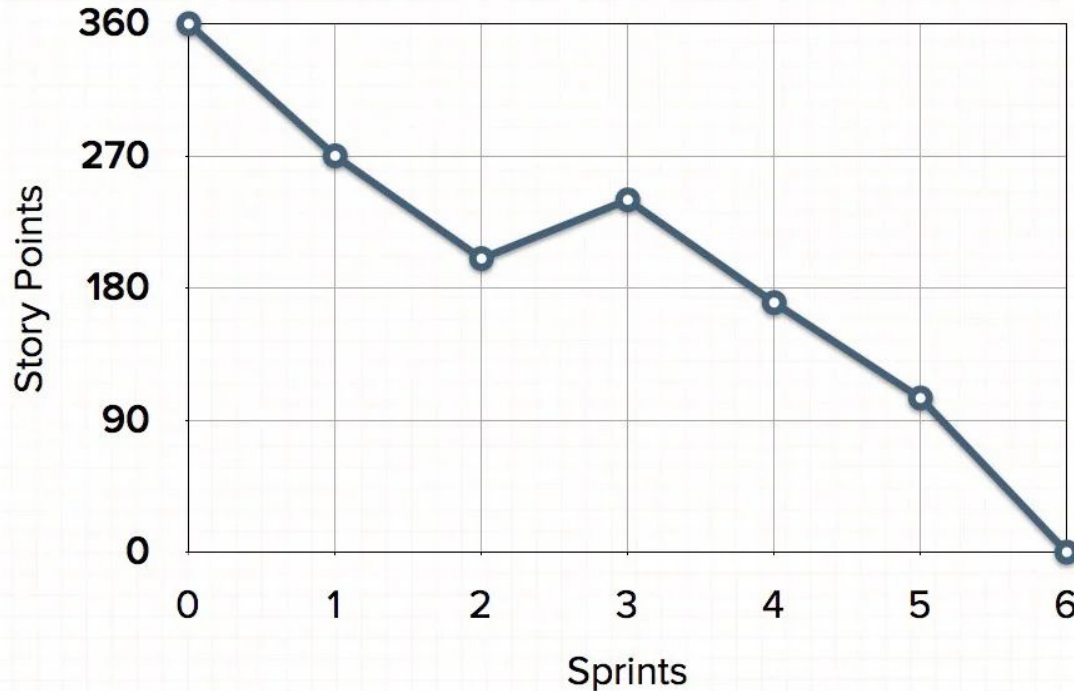


Work completed sooner than anticipated. Over estimated. Should have added more stories.



The look of a typical first sprint – still learning Agile. Failed to complete Sprint.

Release Burndown Chart



The team started a project that was planned to be six sprints. They began with 360 story points of work. To finish within six sprints, they planned to average 60 points per sprint.

Various projective practices upon trending have been used to forecast progress, like burndowns, burn-ups, or cumulative flows. These have proven useful. However, these do not replace the importance of empiricism.

Ultimately, Agile Frameworks like “Scrum” process framework relies on the empirical approach. The empirical process relies on Inspection, adaptation and transparency, enabling continuous feedback loop between development teams and the commercial context.

Outline



**Increment &
DOD**



Sprint Backlog



Sprint Backlog

Case Study : Kopi Kuno

Kanban Board

Product Backlog | 4 ...

Goal : Start selling coffee online

update product

promo feature

payment integration

+

Sprint Backlog | 3 ...

Goal : Getting partners & customer

as a customer i want to add the amount of order so that i can order multiple item

as a customer i want to see order status so that i can get update about my order

+

In Progress (3) | 1 ...

as a customer i want to see the list of coffee menu so that i can choose my order

+

Increment | 4 ...

register customer

register partners

input product

input pricing

+

Sprint #4

Goal : Getting partners & customer

as a customer i want to add the amount of order so that i can order multiple item

5

as a customer i want to see order status so that i can get update about my order

5

as a customer i want to see the list of coffee menu so that i can choose my order

3

Story#4

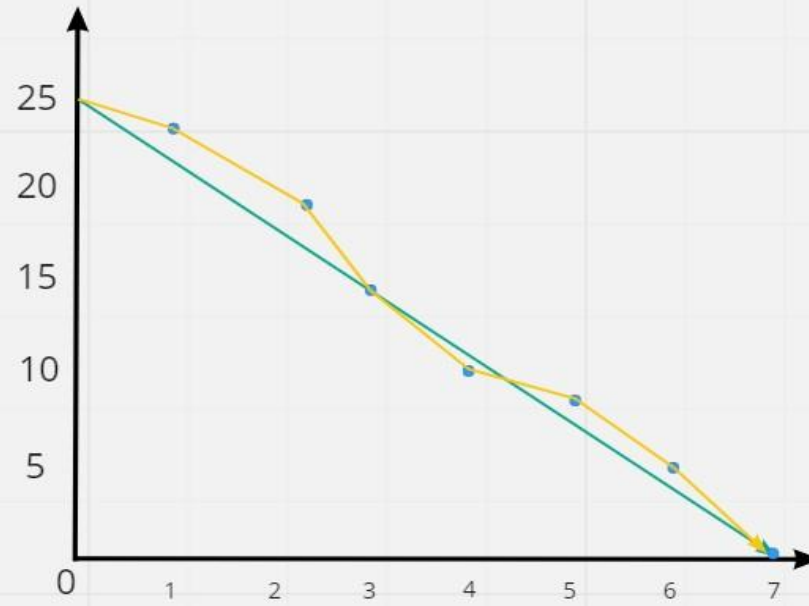
3

Story#6

8

Total Story Point = $5+5+3+3+8 = 24$

Sprint Burndown



Velocity = 24
*managed to finish all

References

- <https://scrumguides.org/docs/scrumguide/v2020/2020-Scrum-Guide-US.pdf>
- <https://scrumguides.org/docs/scrumguide/v2017/2017-Scrum-Guide-US.pdf>
- <https://letsscrumit.com/scrum-artifacts-3-the-increment>
- <https://devhumor.com/media/almost-done-with-software>
- <https://www.scrum.org/resources/blog/definition-done-where-start>
- <https://www.scrum.org/resources/blog/velocity-revolutionary-way-measure-scrum>
- <https://www.scrum.org/resources/blog/myth-velocity-productivity>

Topic 9

Flow Management

Outline

1

Flow Management



2

Kanban Practices



3

Setup Kanban Board

Flow Management

看板

Kan Ban

Kanban is a lean method for defining, managing and improving services that deliver knowledge work
(David J Anderson)

Kanban is a strategy for optimizing the flow of value through a process that uses a visual work-in-progress limited pull system
(The Kanban Guide)

Concept around Kanban

- Kanban manage the system, not the people
- Optimizes flow by improving :
 - Efficiency
 - Effectiveness
 - Predictability



Flow

The movement of value
throughout product
development system

Flow & Empiricism

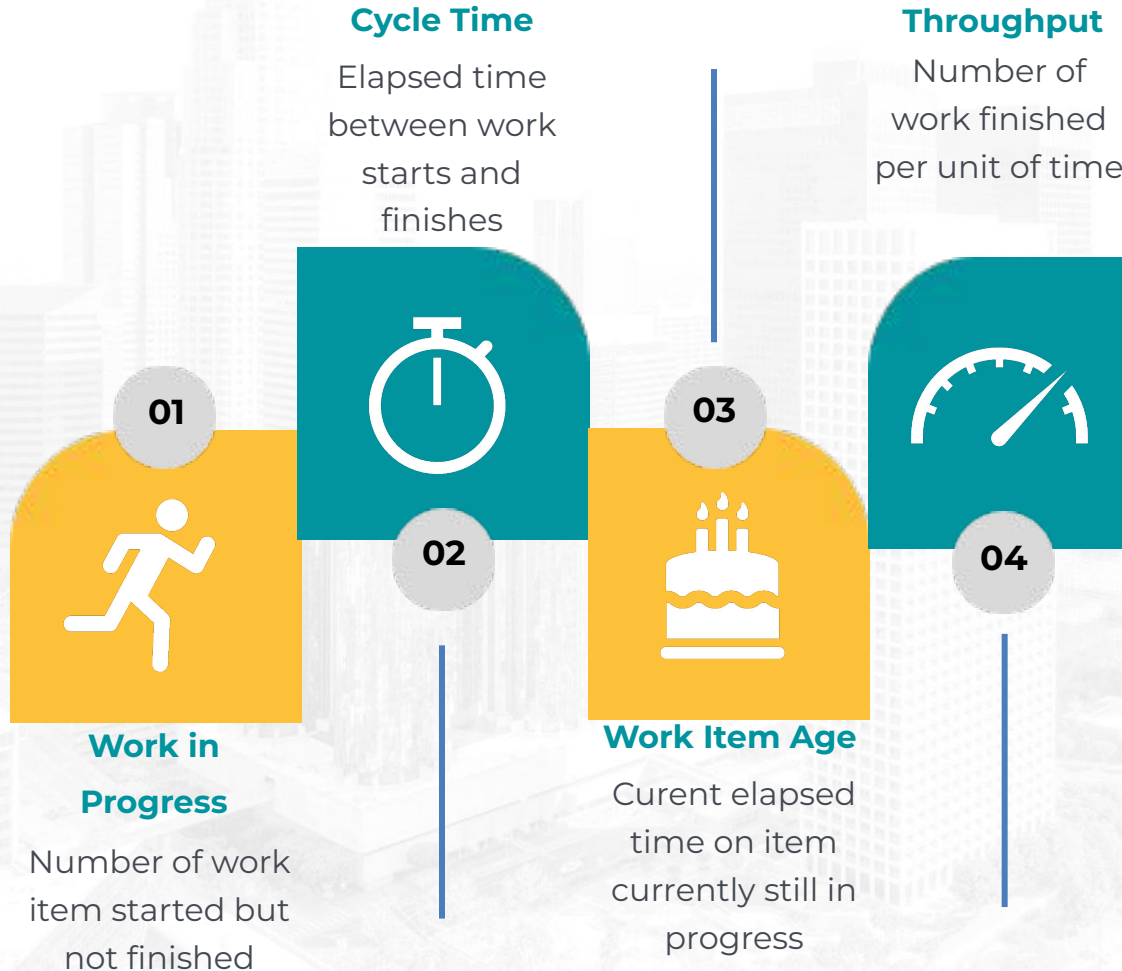
Key to empirical process control is the frequency of the transparency, inspection and adaptation cycle through the feedback loop



Kanban practices provide focus on improving the flow through the feedback loop, optimizing transparency and the frequency of inspection and adaptation for both the product and the process



4 Basic Metrics of Flow



Further Readings

- <https://prokanban.org/the-kanban-guide/>
- <https://resources.kanban.university/kanban-guide/>
- <https://www.scrum.org/resources/kanban-guide-scrum-teams>

Outline



1

Flow Management

2

Kanban Practices

3

Setup Kanban Board

Kanban Practices

Kanban Practices

essential practices in Kanban

Visualize

Define and visualize the workflow

Limit WIP

Limiting work in progress



Manage WIP

Active management of work item in progress

Improve

Inspect, adapt and improve the workflow

Defining & Visualizing The Workflow



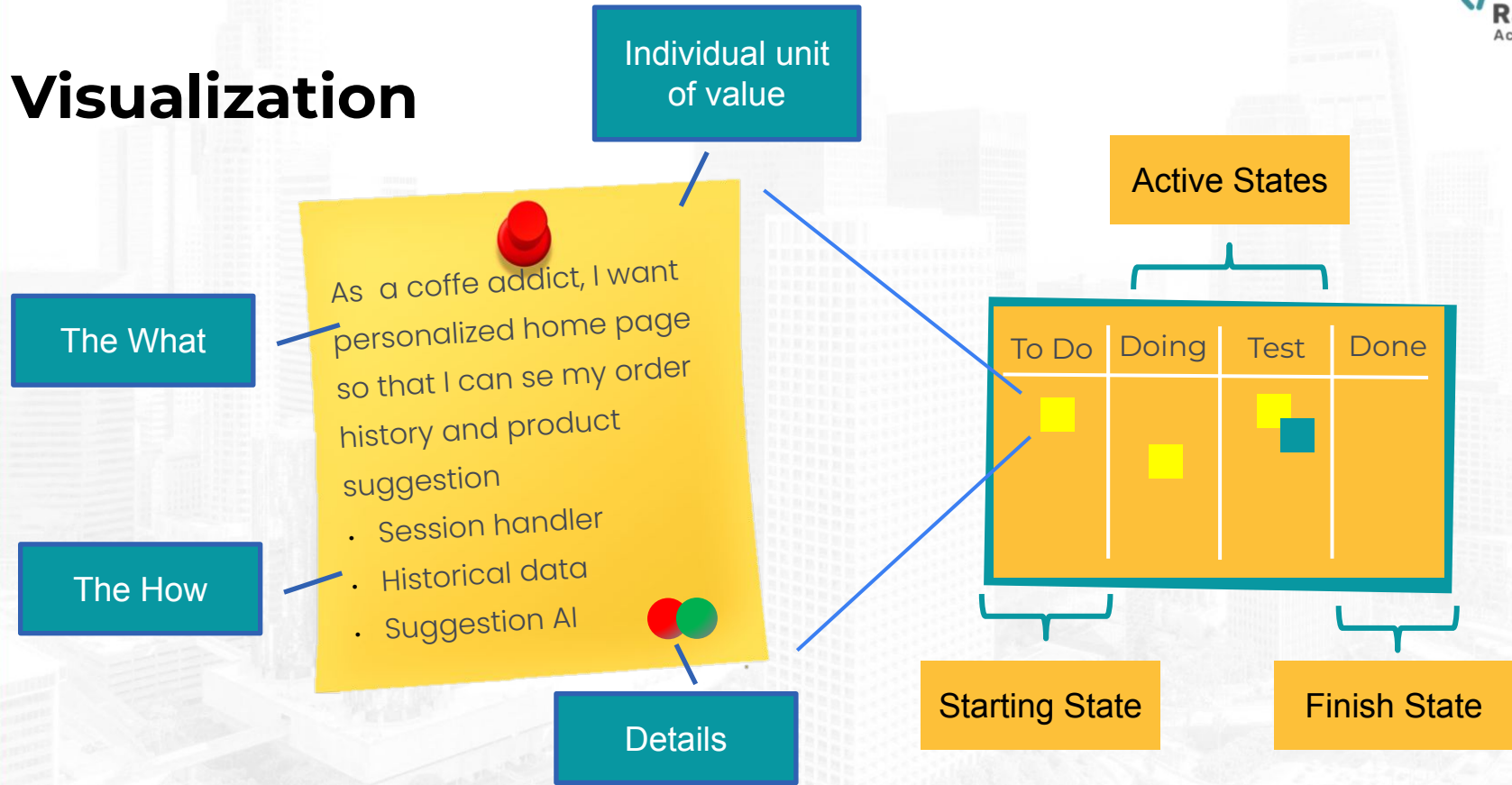
Visualization

- Definition of Work item
- Definition of Workflow

Explicit Policies

- Pull policies
- WIP Limit

Visualization



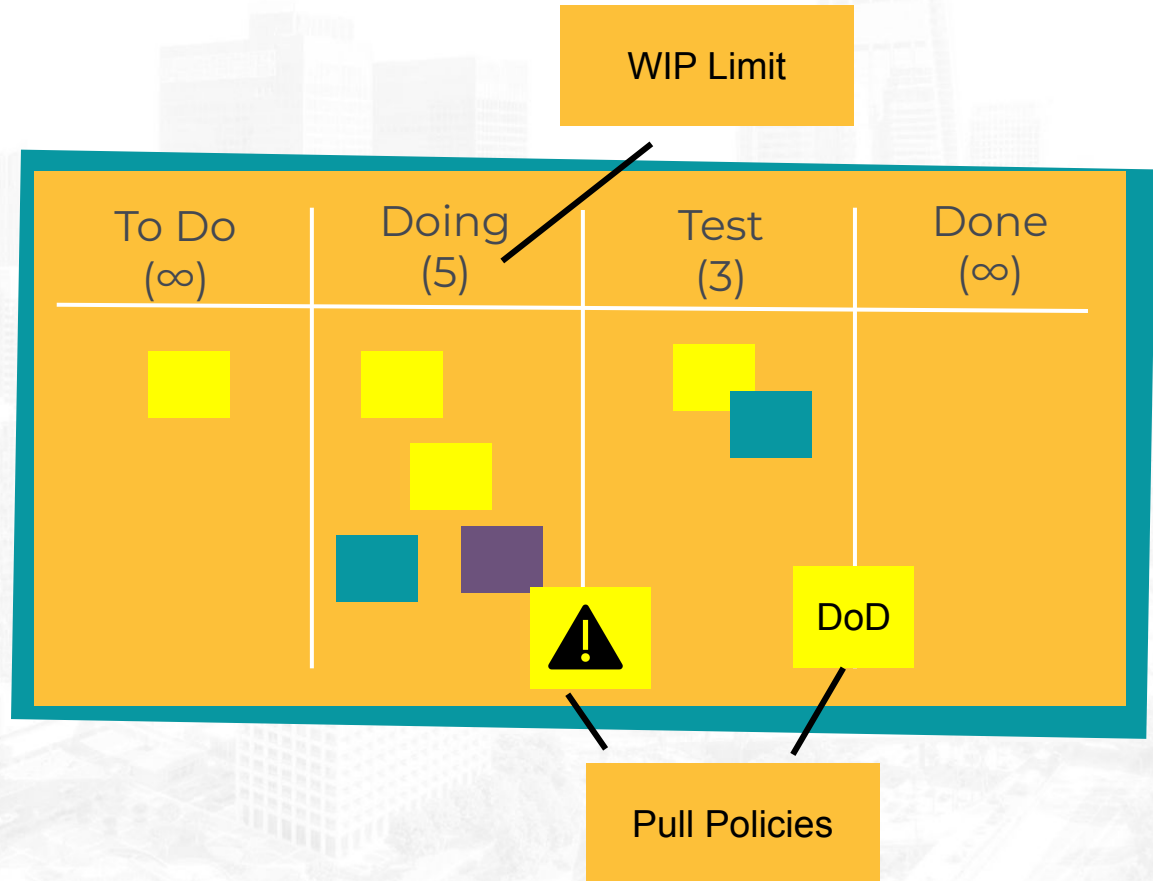
Definition of Work Item

Definition of Workflow

Explicit Policies

WIP Limit : Maximum number of work item in progress, applicable on specific states, per-person or for a whole Kanban system

Pull policies : Specific condition that allow work item to move from one states to another. Different classes of services can have different pull policies



Limiting WIP

Why ?

Creates Pull System



- Promote Self-management and Commitment

Improve Flow

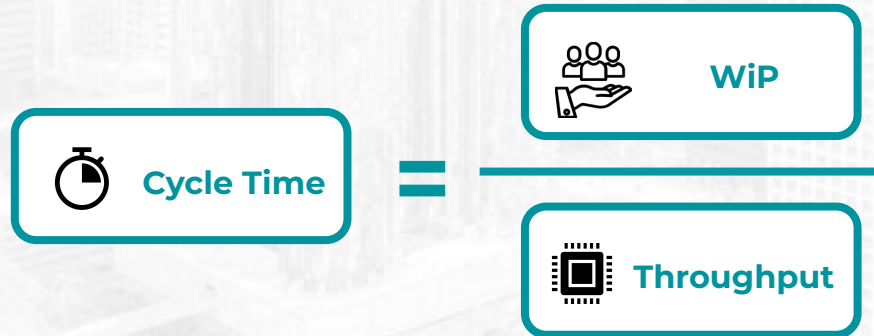


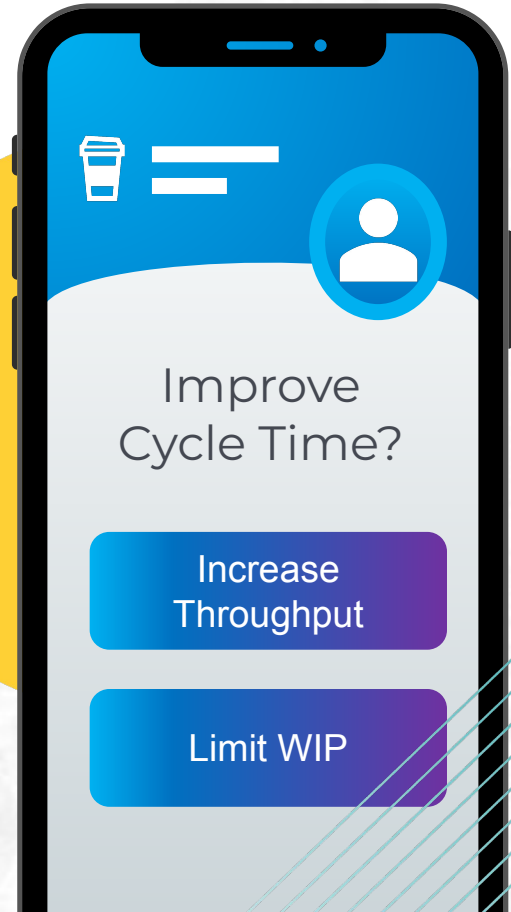
- Increase Focus
- Encourage Communication and Collaboration



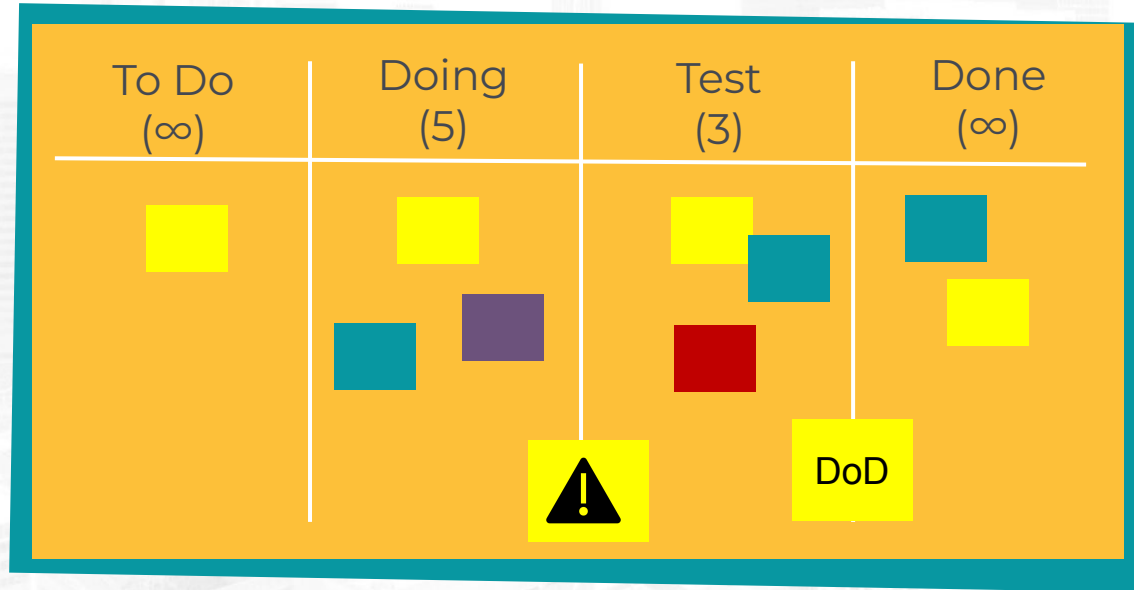
Little's Law

The average cycle time is the average number of work item in progress divided by the delivery rate or throughput of the system

$$\begin{array}{c}
 \text{Cycle Time} = \frac{\text{WIP}}{\text{Throughput}}
 \end{array}$$




Managing Work in Progress



- Stop starting, start finishing, avoid unnecessary aging
- Highlight work item with highest value and/or risk, breaking SLE
- Identify bottleneck and starvation, balancing entry rate with exit rate

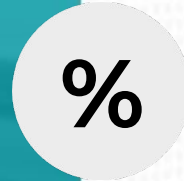
Service Level Expectation

A forecast of how long it should take a given item to flow from start to finish within the workflow
Based on historical data



01

Range of elapsed days or hour

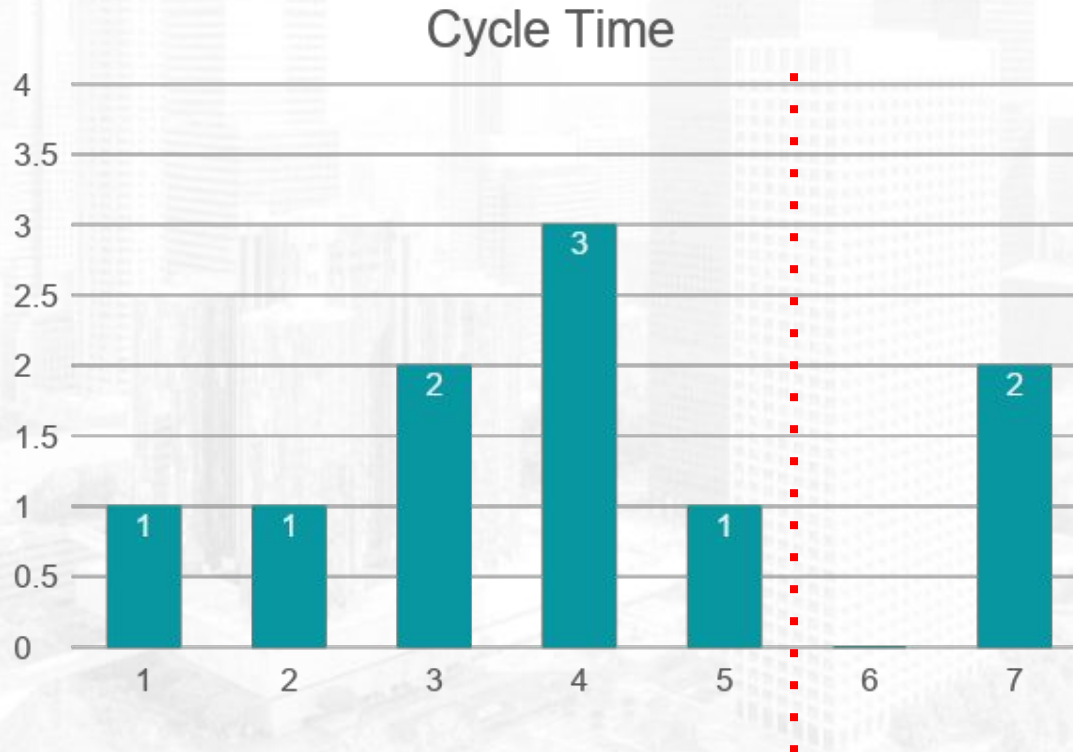


02

Probability associated with that period



SLE Example



Measuring SLE

- Group historical work item with similar cycle time
- Define range
- Calculate percentage within the range

80% work item finished in 5 or less days

Optimizing Work in Progress

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed diam nonummy nibh euismod tincidunt ut laoreet dolore magnaismod tincidunt ut laoreet

1

Balancing WIP limit, too little work and team are underutilized, too much will slow down overall workflow

2

Maintain structure that allow focus and clear out tasks that are finished or unused

3

Some multitasking inevitable, but don't let your team get overwhelmed, be mindful of switching cost

Workflow Inspection & Adaptation

The Product



Value

How valuable for the customer

Upcoming feature and functionality



Cycle Time

How long

How predictable



The Process



Visualization

How clear to understand

How simple to work on



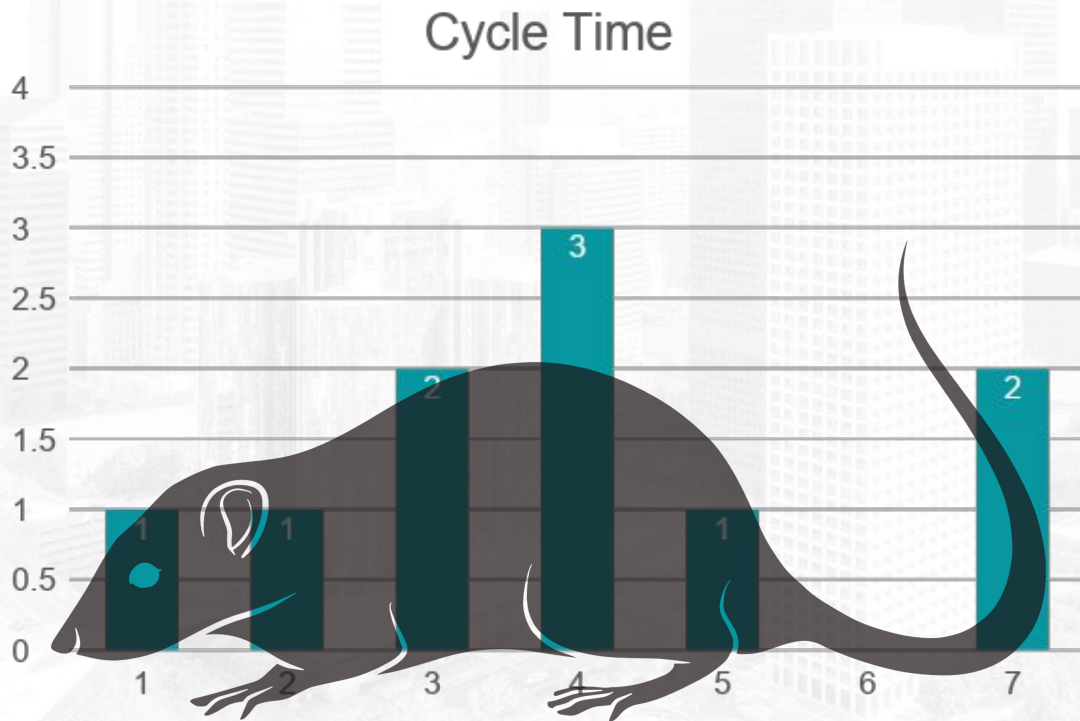
Policies

How effective

How efficient



Improving Cycle Time



Handling the RAT

- Cut the tail, inspect work item that breaking SLE, find unnecessary delay
- Build fat, narrow down the chart, improve predictability
- Move overall cycle time to the left

Further Readings

- <https://prokanban.org/the-kanban-guide/>
- <https://resources.kanban.university/kanban-guide/>
- <https://www.scrum.org/resources/kanban-guide-scrum-teams>
- <https://www.scrum.org/resources/littles-law-professional-scrum-kanban>

Outline



1

Flow Management



2

Kanban Practices



3

Setup Kanban Board

Kanban Board

Scrum with Kanban

Start with what you already are, follow the Scrum guide then apply Kanban practices as evolutionary changes. Improve effectiveness and efficiency while keeping simplicity in mind



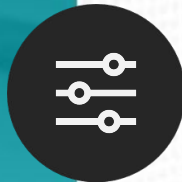
01

Layout your artefacts as starting and finish point



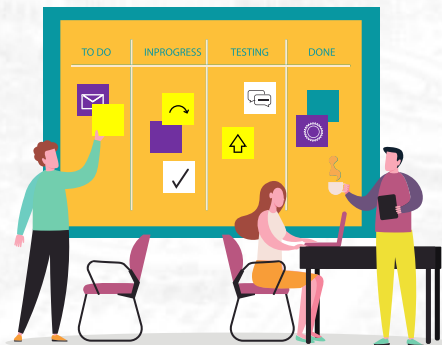
02

Define steps during your active state



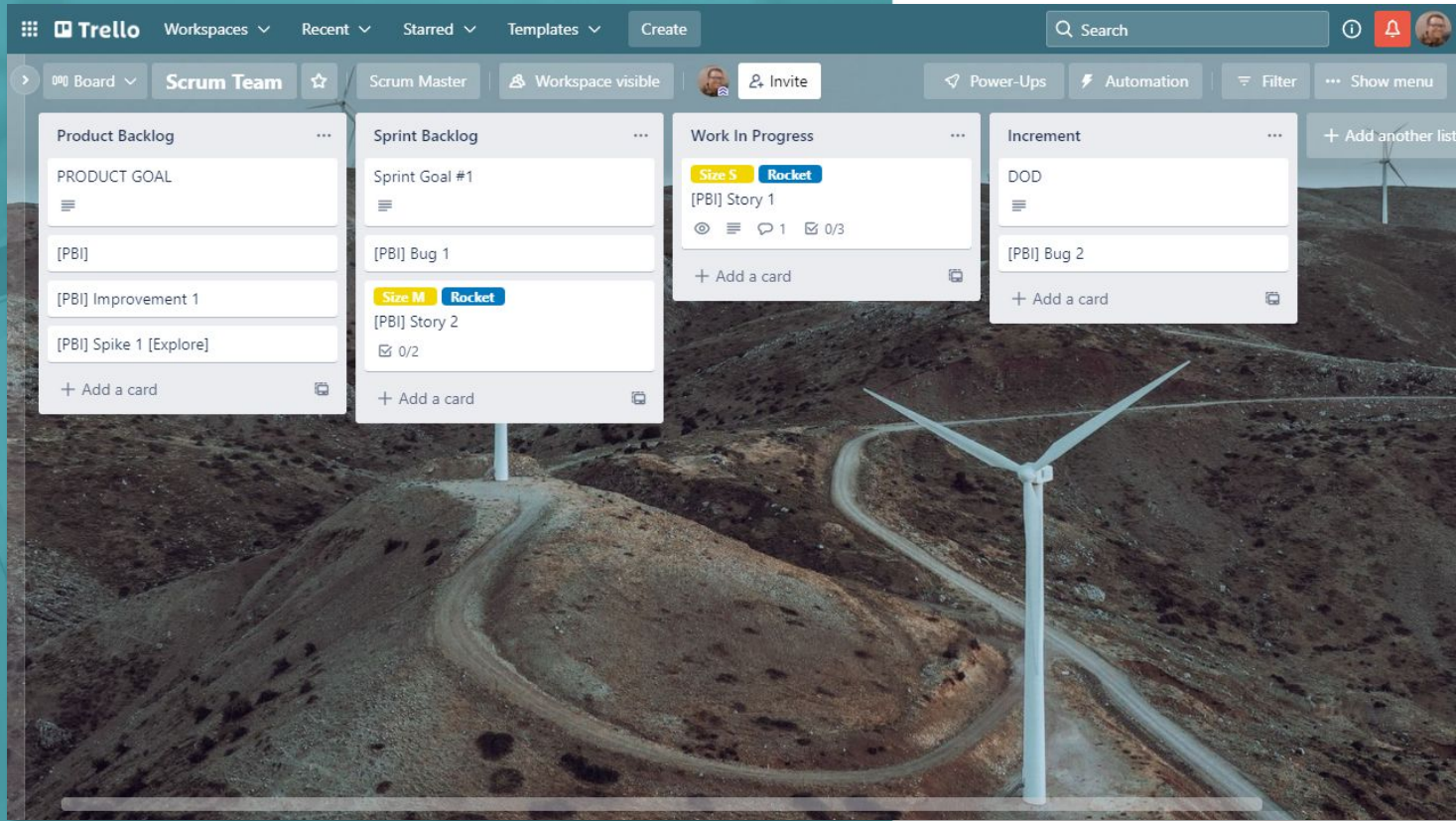
03

Develop basic policies, inspect and adapt along the way



COLOR CODE OF
PROJECTS HERE -->

Physical Kanban Board



Kanban Board
with Trello

Product Backlog | 4

Goal : Start selling coffee online

update product

promo feature

payment integration

Sprint Backlog | 3

Goal : Getting partners & customer

as a customer i want to add the
amount of order so that i can order
multiple item

as a customer i want to see order
status so that i can get update about
my order

In Progress (3) | 1

as a customer i want to see the list of
coffee menu so that i can choose my
order

Increment | 4

register customer

register partners

input product

input pricing

Kanban Board
with Miro

Outline

1

Flow Management



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Terimakasih